SIMATS ENGINEERING

ASSESSMENT TOOL 2: Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 20**

Question Count: 4

Code of Conduct

I Latisha. S (192524071) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Latisha.S

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

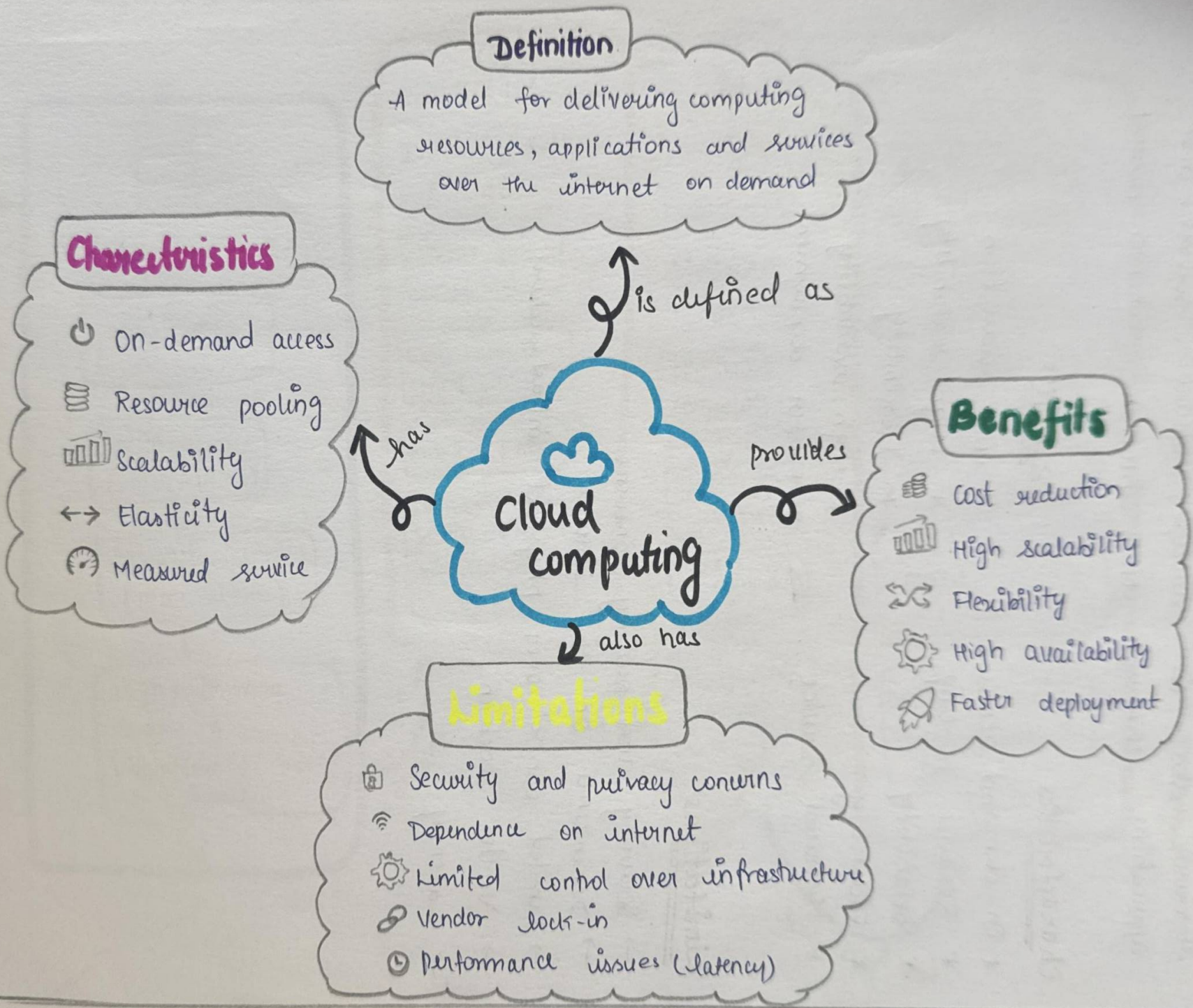
Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

1. CONCEPT MAP 1: Cloud computing fundamentals



1. Definition

Cloud computing means using computing resources such as storage, software, servers, and applications through the internet whenever needed

Characteristics

* On-demand access
* Resource pooling
* Scalability
* Elasticity
* Measured service

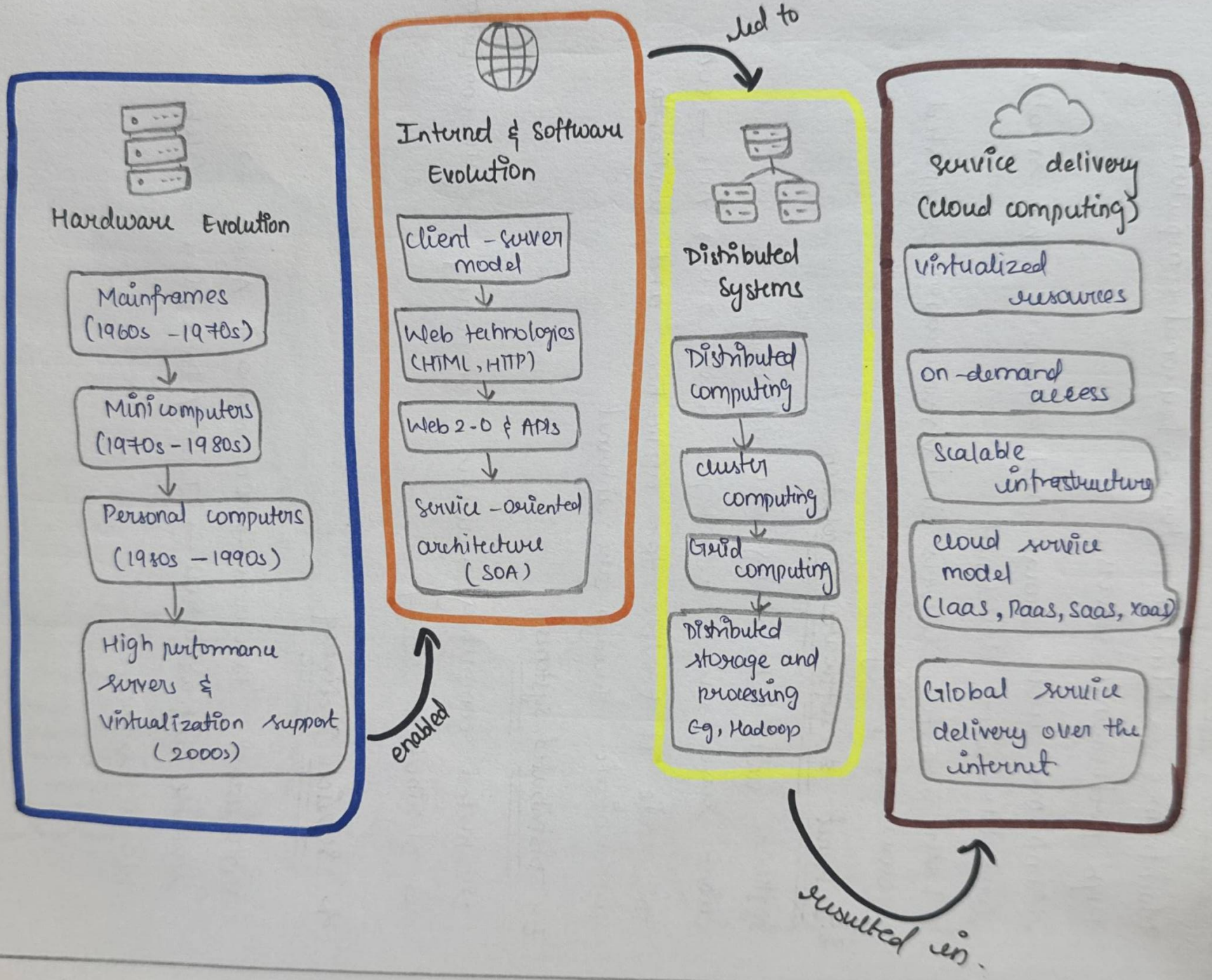
Benefits

* Cost reduction
* High scalability
* Flexibility
* High availability
* Faster deployment

Limitations

→ Security and privacy concerns
→ Dependence on the internet
→ Limited control over physical infrastructure
→ Vendor lock-in
→ Latency

2. CONCEPT MAP 2 : Evolution of cloud computing



The evolution happened through four major stages.

1. Hardware evolution:

Mainframes → Minicomputers → Personal computers → High performance servers.

- Mainframes were large and expensive, then smaller computers were developed
- Later powerful servers and virtualization supported cloud computing

2. Internet & software Evolution

Software and networking also developed.:

client server → Web technologies → web 2.0 & APIs → SOA

This made it possible for applications and services to communicate through the internet

3. Distributed systems

Distributed computing → cluster computing → grid computing → Distributed storage.

4. Service Delivery

Virtualized Resources → On-demand Access → scalable infrastructure → Iaas / Paas / Saas → Global cloud services.

Cloud computing service models

IaaS
(Infrastructure as a service)

provides

Virtualized computing resources

includes

- Virtual machines
- Storage
- Networking
- Load balancers

Provider responsibility
- Virtualization
- Hardware maintainence
- Data center mangement

User responsibility
- OS
- Application
- Data

PaaS
(Platform as a Service)

provides

Platform of developing, running and managing applications

includes

- Runtime environment
- Middleware
- Development tools
- Databases

Provider responsibility
- runtime
- middleware
- Scaling

User responsibility
- Application code
- Data
- Application configuration

SaaS
(Software as a service)

provides

complete software application over the internet

includes

- Email
- CRM
- Office tools
- collaboration software

Provider responsibility
- Entire application
- Infrastructure
- Security
- Updates

User responsibility
- Use the application
- Manage user data
- configure settings

XaaS
(Anything as a service)

provides

Extension of cloud services to anything

includes

- FaaS
- STaaS
- DBaaS
- CaaS
- NaaS

Provider responsibility
- Service management
- Infrastructure
- Availability
- Security

User responsibility
- Use and configure the service
- Manage data

CONCEPT MAP 3: Service Models

Four main cloud services.

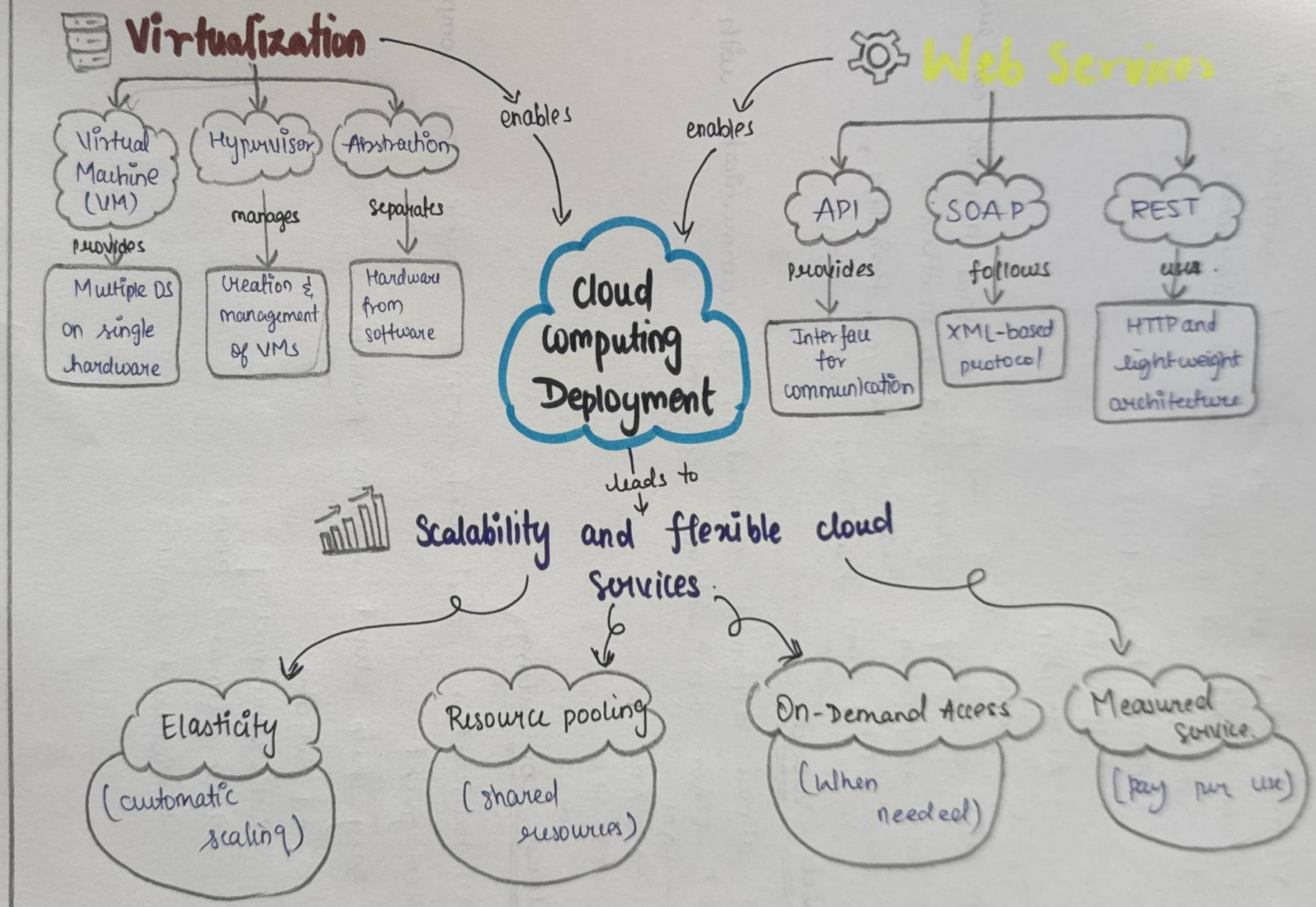
1. Iaas - Infrastructure as a service
   * It provides basic computing infrastructure
   * It includes:
     → Virtual Machine
     → Storage
     → Networking
     → Load balancers

2. Paas - Platform as a service
   * The provider gives a platform for developing and running applications.
   * It includes:
     → Runtime environment
     → Middleware
     → Development tools
     → Databases

3. Saas - Software as a service
   * It gives a complete software applications through the internet
   * It includes
     → Email
     → CRM software
     → Office tools

4. Xaas - Anything as a service.
   * Xaas means almost any IT service can be provided through cloud
   Examples = Faas, STaas, DBaas, Caas, Naas.

CONCEPT MAP 4 : Virtualization and Web Services as Enables of Cloud



Virtualization

* Virtualization allows one physical computer to behave like a multiple virtual computers.
* It involves:
  - Virtual Machine (VM)
  - Hypervisor
  - Abstraction.
* It is useful as a single physical machine can run multiple operating systems and applications.
* This improves resource usage, flexibility, scalability and hardware utilization.

Web services

* It allows different applications to communicate with each other over internet / networks.
  - → API
  - → SOAP
  - → REST

Cloud computing

* Virtualization and web services enable cloud computing deployment
* They help cloud systems provide:
  - → Elasticity
  - → Resource pooling
  - → On-demand access
  - → Measured service.